

Yüksel Kaan Bölükbaş

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Education

Hacettepe University – B.Sc. in Artificial Intelligence Engineering, 4th year Ankara, Turkey Expected 2027

Experience

Baykar Technologies – Intern İstanbul, Turkey August 2026 – September 2026

- Contributed to software development projects in a defense-technology environment using **C++**, working within a professional engineering team on performance-critical systems.

Nart Information Technologies – Intern Ankara, Turkey February 2026 – March 2026

- Built an **hourly trip count forecasting** pipeline for **NYC Yellow Taxi** using **XGBoost, LightGBM, Prophet, and Ridge Regression** (best: $R^2=0.979$); engineered lag features, rolling averages, cyclical encodings, and weather data via Open-Meteo API.
- Processed large-scale **Parquet** files with **DuckDB**; optimized models via **Optuna** hyperparameter tuning with time-series cross-validation.

Datateam Information Technologies – Intern Ankara, Turkey July 2025 – August 2025

- Developed an AI-powered global supplier platform processing **3M+ records** across **25 years** and **42 countries**; implemented automated classification using **PostgreSQL** and **GPT-4o mini**.

Projects

Simurgh Fly – TEKNOFEST International UAV Competition – Team Lead (Captain) Ankara, Turkey February 2026 – Present

- Leading **10-member** multidisciplinary team (Software, Electronics, Mechanics) building an autonomous fixed-wing UAV for AI-driven disaster response; real-time detection via **YOLOv11** on **Raspberry Pi 5 with AI Kit** and crowd density analysis via **DBSCAN**.

Simurgh Fly – TEKNOFEST Fighting UAV (Savaşan İHA) – Software Team Lead Ankara, Turkey August 2025 – February 2026

- Led **10 engineers** developing autonomous combat UAV systems (**Computer Vision, ROS, PX4**); achieved **mAP50=0.96**, **mAP50-95=0.80**, **95% recall** in **Gazebo** simulations.

TEKNOFEST Robotaxi – Autonomous Vehicle – Software Member Ankara, Turkey April 2025 – October 2025

- Conducted target detection model testing and sensor fusion simulations in the **Gazebo** environment.

Simurgh Fly – TEKNOFEST UAV Project – Computer Vision Specialist Ankara, Turkey November 2024 – August 2025

- Trained models on **60,000+ aerial images** achieving **94% accuracy**; optimized inference on **NVIDIA Jetson Xavier NX** via **CUDA** and **TensorRT** to **45+ FPS**.

Operator Audio Monitoring ML Model – Machine Learning Project Remote March 2024 – November 2024

- Built end-to-end **ML pipeline** for real-time operator monitoring via **Audio Signal Processing**: profanity detection, WPM tracking, and aggression detection through frequency spectrum analysis.

Volunteering

Hacettepe University UAV Society – Member of the Supervisory Board Ankara, Turkey August 2026 – Present

- Audited board decisions and financial operations for regulatory compliance; reviewed budgets, expenditure records, and activity reports, presenting findings to the general assembly.

Hacettepe University UAV Society – Vice President Ankara, Turkey February 2026 – August 2026

- Led strategic planning and operational coordination, overseeing **TEKNOFEST** project teams and representing the society in university-level activities.

Hacettepe University UAV Society – Director, Project Coordination Unit Ankara, Turkey August 2025 – February 2026

- Supervised multiple **TEKNOFEST teams** and coordinated member selection processes, ensuring high-performance standards across the society.

Technical Skills

Languages & Frameworks: Python, Java, C++, Machine Learning, PyTorch, YOLO, Computer Vision, PostgreSQL, ROS, Streamlit, Linux, CUDA, TensorRT, Git.